1 Notation and conventions

Indices and sets. We use lower-case Latin letters for indices and calligraphic upper-case for sets:

- $r \in \mathcal{R}$ — receiver. Single-receiver modes (PPP, PPP-RTK) have $|\mathcal{R}| = 1$; baseline modes have a fixed reference and one or more rovers; the network producer (`PPP_RTK_Network`) carries a peer set of $|\mathcal{R}| \geq 2$ stations (50–305 in practice) with no reference/rover asymmetry.
- $g \in \mathcal{G} = \{\text{GPS}, \text{GAL}, \text{BDS}, \text{GLO}, \text{QZS}, \dots\}$ — constellation.
- $i \in \mathcal{S}_g$ — satellite of constellation g. p_g denotes the pivot satellite of constellation g under the BSSD parameterisation (Section 4).
- $s \in \mathcal{B}$ — signal/observation type, identified by its RINEX-3 three-character code (e.g. `C1C`, `L2W`, `L5X`).
- $k \in \mathbb{N}$ — discrete epoch index.

Vectors and matrices. Bold lower-case for vectors ($\boldsymbol{x}$, $\boldsymbol{r}$); bold upper-case for matrices ($\boldsymbol{H}$, $\boldsymbol{P}$). $\boldsymbol{a}^{\mathrm{T}}$ is the transpose. A hat denotes a state estimate; a tilde a one-step prediction: $\hat{\boldsymbol{x}}_{k|k}$ vs. $\tilde{\boldsymbol{x}}_{k|k-1}$.

Observables. For each receiver/satellite/signal triple, the engine consumes two real-valued observables (in metres) per epoch:

- $P^i_{r,s}$ — pseudorange (code), in metres.
- $\Phi^i_{r,s}$ — carrier phase, scaled to metres via λ_s. Beyond the range, clock, and atmosphere terms it shares with the code, its ambiguous (non-range) part is $\lambda_s N^i_{r,s}$ plus the fractional phase biases $b_{r,s} - b^i_s$ (Equation (5)).

Where the distinction matters, s may further denote the band ($L_1, L_2, L_5, \dots$) and the tracking attribute (e.g. `C`, `W`, `X`). The leading observable-type letter (`C`/`L`) selects code vs. phase; the frequency-only quantities f_s, λ_s, and $N^i_{r,s}$ depend only on the (band, attribute) part, so a code and its sibling phase share them. This matches the state keys, which carry the full phase code for ambiguities (`"G01_L1C"`) but only the band digit for receiver code biases (`"GPS_IFB2"`).

Frequencies and wavelengths. f_s is the carrier frequency (Hz) and $\lambda_s = c/f_s$ the wavelength (m). The frequency-squared scaling factor used to project the L1-equivalent slant ionospheric delay onto signal s is

$$\mu_s \;:=\; \frac{f_1^2}{f_s^2}. \tag{1}$$

Here f_1 is the GPS L1 frequency (1575.42 MHz), used as the common reference for *every* constellation — not the band-1 frequency of the satellite's own system. The estimated state $I^i_{r,1}$ is therefore GPS-L1-equivalent even for BDS/GLO/GAL satellites (`MeasurementModel::buildJacobianRow`: `f_ref = Constants::GPS_L1_FREQ`).

Differencing operators. Three differencing operations recur throughout the document:

$$\Delta q^{i,p_g}_{r,s} := q^i_{r,s} - q^{p_g}_{r,s}, \quad \nabla q^i_{r_1 r_2,s} := q^i_{r_1,s} - q^i_{r_2,s}, \quad \nabla\Delta q^{i,p_g}_{r_1 r_2,s} := \Delta\big(\nabla q^i_{r_1 r_2,s}\big).$$

Read Δq as "between-satellite single difference of q relative to pivot p_g," ∇q as "between-receiver single difference," and $\nabla\Delta q$ as "double difference."

Sign conventions. We adopt the convention that the slant ionospheric delay $I^i_{r,s}$ enters *additively* on the code observable and *subtractively* on the carrier phase observable; this is the source of the well-known "code–phase divergence." All other range-additive terms (geometry, clocks, troposphere, biases) carry the same sign on both observables. See Equations (4) and (5) for the explicit equations.

Symbol table.

Symbol	Units	Meaning
$\boldsymbol{r}_r, \boldsymbol{r}^i$	m	ECEF position of receiver/satellite
ρ^i_r	m	geometric range $\|\boldsymbol{r}^i - \boldsymbol{r}_r\|$
$\boldsymbol{e}^i_r$	—	unit LOS from r to i in ECEF
$c\,\mathrm{d}t_r$	m	receiver clock offset (scaled by c)
$c\,\mathrm{d}t^i$	m	satellite clock offset (CLK product in user modes; estimated state `"<sat>|SC"` in `PPP_RTK_Network`)
$c\,\delta t^{(g)}_r$	m	receiver inter-system time offset for constellation $g \neq g_{\mathrm{ref}}$
T^i_r	m	total tropospheric path delay
$\mathrm{ZHD}^i_r, \mathrm{ZWD}^i_r$	m	zenith hydrostatic / wet components
$I^i_{r,1}$	m	L1-equivalent slant ionospheric delay
$N^i_{r,s}$	cyc	integer carrier-phase ambiguity
$b_{r,s}, b^i_s$	m	receiver / satellite phase biases
$\bar{b}^i_s$	m	estimable satellite phase bias (network mode; Section 2)
$B_{r,s}, B^i_s$	m	receiver / satellite code biases (OSB)
$B^{(g,b)}_{r,k}$	m	per-FCN-slot GLONASS receiver IFB, band b; here $k \in [-7, +6]$ is the signed FDMA channel number (a context-local reuse, not the epoch index)
$\varepsilon^P, \varepsilon^\Phi$	m	residual (multipath + thermal)
$\boldsymbol{v}_r$	m/s	receiver velocity (optional / IMU state)
$\lambda_{\mathrm{WL}},\ \lambda_{\mathrm{NL}},\ \lambda_{\mathrm{EWL}}$	m	wide- / narrow- / extra-wide-lane wavelength $c/(f_a \mp f_b)$ for the lane's band pair
λ_{IF}	m	effective ionosphere-free wavelength $\|c_1\lambda_1 - c_2\lambda_2\|$ with the IF coefficients $c_1 - c_2 = 1$; not an integer wavelength (the cycle-slip document)
$\boldsymbol{C}^e_b$	—	body-to-ECEF attitude (direction-cosine) matrix

2 State vector layout

The Kalman filter operates on a state vector $\boldsymbol{x}_k \in \mathbb{R}^{n_k}$ whose dimension n_k *varies across epochs*: per-satellite and per-(satellite, signal) substates are added when a new sat/signal is first observed and pruned (or marked stale) when the satellite drops below the elevation mask or its phase track is interrupted by a cycle slip without recovery. All bookkeeping — key→index maps, free-list re-use, and dimensional reshaping of $\boldsymbol{x}$ and $\boldsymbol{P}$ — is encapsulated in `StateManager`; `KalmanFilter` consumes the state through that interface and never indexes raw memory by hard-coded offsets.

2.1 Schematic layout

In schematic form, suppressing the epoch index k:

$$\begin{aligned}\boldsymbol{x} \;=\; \big[\; & \underbrace{\boldsymbol{r}_r^{\mathrm{T}}}_{3},\ \underbrace{\boldsymbol{v}_r^{\mathrm{T}}}_{0\text{ or }3},\ \underbrace{c\,\mathrm{d}t_r}_{1},\ \underbrace{(c\,\delta t_r^{(g)})^{\mathrm{T}}_{g\neq g_{\mathrm{ref}}}}_{|\mathcal{G}|-1},\\ & \underbrace{\mathrm{ZWD}_r,\ (g_N, g_E)}_{1+0\text{ or }1+2},\ \underbrace{(c\,\mathrm{d}t_r^{\phi})}_{0\text{ or }1},\\ & \underbrace{(I^i_{r,1})^{\mathrm{T}}_{i\in\mathcal{V}_k}}_{|\mathcal{V}_k|},\ \underbrace{(N^i_{r,s})^{\mathrm{T}}_{(i,s)\in\mathcal{A}_k}}_{|\mathcal{A}_k|},\\ & \underbrace{(B_{r,s})^{\mathrm{T}}_{s\in\mathcal{B}}}_{|\mathcal{B}|},\ \underbrace{(B^i_s)^{\mathrm{T}}_{(i,s)\in\mathcal{D}_k}}_{|\mathcal{D}_k|}\;\big]^{\mathrm{T}}. \end{aligned} \tag{2}$$

where

- $\mathcal{V}_k$ is the set of satellites currently visible at epoch k (i.e. above the elevation mask, with valid orbit/clock products and at least one tracked signal),
- $\mathcal{A}_k \subseteq \mathcal{V}_k \times \mathcal{B}$ is the set of phase-tracked (satellite, signal) pairs,
- $\mathcal{D}_k \subseteq \mathcal{V}_k \times \mathcal{B}$ is the subset of (satellite, signal) pairs carrying a satellite code-bias state: the pairs for which the BIA product lacks a satellite OSB or — under the network producer's satellite-OSB mode — every raw code pair (see `estimation.estimate_sat_biases` and `network.estimate_sat_code_bias`).

For brevity the schematic suppresses several conditional refinements detailed below: the rate states that accompany velocity (a receiver clock-rate and one inter-system clock-rate offset per non-reference constellation), the optional per-satellite slant-iono *rate* companion under `enable_iono_rate`, the network-producer satellite-clock and receiver-phase-bias states, the nine-state inertial augmentation of the inertial-aiding document, and the fact that the entire receiver-side clock and inter-system-offset block is removed under the BSSD parameterisation (Section 4.1).

2.2 Component-by-component description

Position $\boldsymbol{r}_r \in \mathbb{R}^3$. ECEF Cartesian receiver position. Always present. Initialised either from the RINEX header approximate position or, after the first epoch, from a Weighted Least-Squares (WLS) seed solution to within the linearisation radius of the geometric range model (Section 3.3).

Velocity $\boldsymbol{v}_r \in \mathbb{R}^3$. Optional. Enabled by `estimation.estimate_velocity = true`, or forced on by tightly-coupled IMU integration. When present, position and velocity form a coupled continuous-white-noise-acceleration (CWNA, integrated-random-walk) block driven by a *single* acceleration spectral density `process_noise.acceleration_psd_q` (the CWNA process-noise model in the estimation document); the scalar `position_q` of the velocity-free case is then unused. Enabling velocity also introduces a receiver clock-rate state $c\,\dot{\mathrm{d}}t_r$, and one inter-system clock-rate offset per non-reference constellation, each CWNA-coupled to its bias counterpart (`clock_rate_q`, `system_time_rate_offset_q`).

Receiver clock $c\,dt_r$. Single state, in metres. Process noise is large (`clock_bias_q` $\sim 10^4\,\mathrm{m}^2$ per second) so the state is effectively re-estimated each epoch; this captures oscillator drift without imposing a stochastic model that may not match the receiver's actual oscillator. Under the BSSD (between-satellite single-difference) parameterisation (Section 4.1) the receiver clock is unobservable and the state is dropped entirely — as are the inter-system offsets below — so the whole receiver-side clock block vanishes in PPP-RTK SD mode.

Inter-system time offsets $c\,\delta t_r^{(g)}$. One per active constellation other than the chosen reference (default GPS). These absorb the per-constellation clock biases that arise from mixing analysis-centre-defined clock products across systems ("GAL − GPS time offset," etc.).

Tropospheric ZWD and gradients. ZWD_r models the zenith wet delay; the zenith hydrostatic component ZHD_r is treated deterministically via the Saastamoinen model [3]. Total slant tropospheric delay is then

$$T_r^i \;=\; m_h(\theta_r^i)\,\mathrm{ZHD}_r \;+\; m_w(\theta_r^i)\,\mathrm{ZWD}_r \;+\; m_w(\theta_r^i)\cot\theta_r^i\big[g_N\cos\alpha_r^i + g_E\sin\alpha_r^i\big], \tag{3}$$

with elevation θ_r^i, azimuth α_r^i, and the Niell or VMF mapping functions m_h, m_w. Gradients (g_N, g_E) are optional.

Per-satellite slant ionosphere $I_{r,1}^i$. PPP-RTK-only. One state per visible satellite, expressed at the L1 frequency; signal-specific delays follow Equation (1). Initial value seeded from the Klobuchar broadcast model when no external ionospheric product is available. Its process noise is a random walk whose zenith spectral density `ionosphere_q` is projected onto the slant by the squared obliquity factor $1/\sin^2\theta_r^i$ (the ionospheric process-noise model in the estimation document), letting low-elevation states vary faster. Modelling these per-sat states explicitly is what permits PPP-RTK to retain integer ambiguity recovery without forming the ionosphere-free combination (which amplifies code noise by a factor of ~ 3). Optionally, under `estimation.enable_iono_rate` (default off), each visible satellite carries a companion constant-velocity block $[\,I_{r,1}^i,\ \dot{I}_{r,1}^i\,]$: the slant-iono *rate* $\dot{I}_{r,1}^i$ is unobserved directly ($H = 0$) and driven only by the F-coupling into the level state, with its own spectral density `process_noise.iono_rate_psd` and its own slip response (`reinflateIonosphereRateVariance`), letting the filter track a linear slant ramp rather than a pure random walk.

Carrier-phase ambiguities $N_{r,s}^i$. One state per phase-tracked (i, s) pair, in metres (the state value is λN; the `nl_wl` basis below is the exception, carrying integers in cycles). Treated as float during the prediction/update cycle and resolved to integers post-hoc by the LAMBDA-family algorithms in `filter/AmbiguityResolver` [5] when AR is enabled. Cycle slips trigger a re-seed via `StateManager::resetAmbiguity`, which *decorrelates* the state rather than repairing it: the ambiguity's covariance row and column are zeroed and its diagonal reset to σ_m^2, while the state *value* is left unchanged (it is overwritten by the geometry-based seed in `ensureRequiredStates`). This follows the project's reset-not-repair convention — a slipped ambiguity carries no trustworthy correlation to the rest of the state, so the cross-state correlations are broken, not preserved.

Receiver code biases $B_{r,s}$. These absorb the receiver-side inter-frequency biases that the OSB convention does not reference to zero. They are *not* one per signal: `StateInitializer::resolveReceiverBias` keys them per (system, frequency band) as inter-frequency biases

("<system>_IFB<band>") plus, in the alternative scheme, a single per-system differential code bias ("<system>_DCB"). Like the receiver clock they are receiver-common and so are suppressed under BSSD. On the network path these columns have no Kalman time update at all — the producer runs no `KalmanFilter` (the filtering document) — so they are re-estimated afresh each epoch unless `network.carry_rx_code_bias` is armed: an opt-in, default-off diagonal random-walk carry ($\mathrm{var} + q\,\Delta t$) held in the network run context rather than in `StateManager`, guarded at load by the requirement of a time-linked code datum (`network.code_datum_time_linked`, absent which the loader forces the carry off) and at inject time by an innovation gate (`rxCbInjectGate` in `filter/NetworkRunTypes`) that withholds the fold when the carried level and the station's own estimate disagree at receiver-reset scale. It is not yet field-validated.

Per-slot GLONASS inter-frequency bias $B_{r,k}^{(g,b)}$. Optional, off by default. GLONASS FDMA assigns each slot $k \in [-7, +6]$ a distinct carrier frequency $f = f_{\mathrm{base}} + k\,\Delta f$, so the receiver group delay differs *per slot* and a single per-band IFB state cannot represent it. When `estimation.glonass_ifb.enable = true` (and the prerequisite `estimation.estimate_ifb`; the loader warns and the flag is inert otherwise), `resolveReceiverBias` instead emits one state per (FDMA band $b \in \{\mathrm{G1}, \mathrm{G2}\}$, slot k) with unit measurement coefficient, keyed "GLONASS_IFB<band>_k<±slot>". The central channel $k = 0$ is the held datum and produces no state; its constant is absorbed by the GLONASS inter-system time offset. CDMA bands (G3/G4/G6) have a slot-independent frequency and retain the ordinary per-band treatment above. Crucially, unlike the receiver-common biases these columns *survive* between-satellite single differencing — two GLONASS satellites on different slots do not cancel — so `BssdTransform` and `StateManager` special-case them (via `isGlonassPerSlotIfbKey`) to exempt them from the receiver-block zeroing; only the inter-slot *differences* remain estimable in SD.

Satellite code biases B_s^i. Per (satellite, signal) pair, but *only when the BIA file does not supply an OSB* for that pair — typically a third-frequency signal in a triple-frequency setup. That fallback rule is the user-path and network-V1 behaviour. Under the network producer's satellite-OSB mode (`network.estimate_sat_code_bias`, also implied and forced on by `network.estimate_iono_sh`) the loader instead *promotes* the state to every raw code (i, s) rather than only the uncovered pairs, and suppresses the external `.BIA` entirely so the network's own estimate is the single code-bias source; the datum that makes it estimable is derived in the network-mode document. When present, these states are weakly stochastic (`sat_bias_q` $\sim 10^{-10}$) since the underlying physical biases are nearly stationary on the daily timescale.

Satellite phase biases $\bar{b}_s^i$ (network mode). Present only in the network producer (`ProcessingMode::PPP_RTK_Network`) and gated by `estimation.estimate_sat_phase_biases`. The JSON default is off and the flag is inert in every other mode, but this is not an experimental opt-in: in the one mode where the states exist the network-producer loader force-enables it unconditionally (the *forcing estimate_sat_clocks + estimate_sat_phase_biases* branch in `GnssConfigLoadNetwork`), and the mirror `GnssConfig::applyNetworkConsumerCouplings` turns it back off for the closed-loop consumer, which *consumes* these states rather than estimating them. The user side imports the satellite phase bias as a known SSR correction (the SSR-corrections document) or absorbs it into the float ambiguity; the network instead carries an *explicit* per-(satellite, signal) phase-bias state on the undifferenced phase rows (the estimable bias of Odijk [2], lumping the network pivot-receiver ambiguity). On a single station's row it is co-linear with the per-receiver float ambiguity and is separable only *across* stations in the joint network estimator

(the network-mode document). It lives in a *distinct* state map from the satellite *code* biases above (`StateManager::sat_phase_bias_indices_`), keyed `"<sat>|<signal>|PB"`, and is near-stationary between epochs at the rate `sat_phase_bias_q` — which on the network path is applied as the inflation of a shared-border *carry prior*, not as a Kalman $\boldsymbol{Q}(\Delta t)$ entry (see the satellite-clock paragraph below).

Receiver phase biases $\delta_{r,s}$ (network mode). The receiver-side counterpart of the satellite phase bias, present only in the network producer and gated by `network.phase_datum_eq34` (the Eq-34 / single-receiver-chain phase datum; see the network-mode chapter). Giving the receiver fractional phase bias its *own* estimable state $\delta_{r,s}$ stops it lumping into the emitted satellite phase bias, so the emitted between-satellite reference stays integer-consistent across *all* satellites (each chain hop then contributes only an integer). It is per (receiver, system, signal), keyed `"<system>_RPB_<signal>"` in its own map (`StateManager::receiver_phase_bias_indices_`). Structurally it is unlike every other bias block described here: these are *local* states, each observed by a single station. Whether they carry across epochs is a knob. At `network.phase_datum_eq34_rpb_q_m2s` $= 0$ each $\delta_{r,s}$ is re-estimated fresh every epoch (the original Card-3 behaviour); at the non-zero default the producer folds a local time-linking prior toward the previous estimate with variance var $+ q\,\Delta t$, which suppresses the per-epoch bounce that an only weakly observable local state otherwise injects, anti-correlated, into every individual ambiguity float. That carry lives *outside* `StateManager` — in the network run context (the `rpb_carry` map of `app/NetworkRunContext`), read during the parallel block assembly, harvested post-solve, and erased for any (i, s) whose chain frame stepped at that epoch's election (`filter/NetworkChainDatum`). It is forced off under `network.carry_invariant`, where the loader warns that the q is inert. Either way the receiver-side wander sinks into $\delta_{r,s}$ while the time-linked integer information stays in the ambiguity carry.

Satellite clocks $c\,\Delta\delta t^i$ (network mode). Also network-producer-only, gated by `estimation.estimate_sat_clocks` — like the phase bias above, off by JSON default and inert elsewhere, but force-enabled unconditionally by the network-producer loader at the same site. The network estimates an *explicit* per-satellite clock correction $\Delta\delta t^i$ (in metres, relative to the a-priori precise/real-time product clock already removed from `obs.modeled`) so the emitted clocks stay consistent with the chosen S-system datum. It is *per satellite*, not per signal — one state per satellite, with no signal field — and the satellite *orbit* stays fixed from the product; only the clock is estimated. It lives in a distinct map (`StateManager::sat_clock_indices_`), keyed `"<sat>|SC"`. Its cross-epoch rate `sat_clock_q` is near-white, but on the network path it is *not* a Kalman $\boldsymbol{Q}(\Delta t)$ entry: the producer runs no `KalmanFilter` (the filtering document), and the rate instead inflates the shared-border carry prior as var $+ q\,\Delta t$. The same holds for `sat_phase_bias_q` and `sat_code_bias_q`; all three are authorable either through their legacy `filter.process_noise.*` keys or through the network-scoped `network.process_noise` class table, whose per-class fallbacks equal those legacy defaults. Between full solves the emitted clock may additionally be refreshed by the split-cadence fast tier (`network.fast_clock`, default off and currently gate-failed); that tier estimates a clock *delta* in its own least-squares tick and creates no state-vector row, so it does not appear here — see the network-mode document.

Inertial augmentation. When a tightly-coupled IMU is active the core state grows by nine further entries — a three-axis attitude error and the accelerometer and gyroscope biases — propagated by the strapdown mechanisation rather than the random-walk process model. These are treated in full in the inertial-aiding document.

Baseline differential troposphere. In baseline mode the per-station tropospheric, clock, ionosphere, and bias states are absent; an optional single differential zenith-wet-delay state $\Delta\mathrm{ZWD}_{r_{\mathrm{rov}}r_{\mathrm{ref}}}$ may be carried (`baseline.estimate_differential_tropo`, Section 4.3).

2.3 Dynamic indexing

The variable-dimension blocks are addressed via key→index maps maintained by `StateManager`. The canonical key formats are defined once in `include/filter/StateKeySchema.hpp`, each with a `make`/`parse` pair:

- iono key: `"<sys><prn>"`, e.g. `"G05"` (delimiter-free);
- ambiguity key: `"<sys><prn>_<signal>"`, e.g. `"G05_L1C"` — split on the *first* underscore, so an iono-free combination `"G05_L1C_L5Q"` retains its compound signal;
- satellite signal-bias key (code): `"<sys><prn>|<signal>"`, e.g. `"G05|C1W"` — a *pipe* delimiter and a distinct map, *not* the underscore ambiguity namespace;
- satellite phase-bias key (network): `"<sys><prn>|<signal>|PB"`, e.g. `"G05|L1C|PB"` — the trailing `|PB` keeps it disjoint from the code-bias namespace, in its own map (`sat_phase_bias_indices_`);
- receiver phase-bias key (network): `"<system>_RPB_<signal>"`, e.g. `"Galileo_RPB_L5Q"` — the `_RPB_` infix keeps it disjoint from the receiver code-bias namespace (which matches on `IFB`/`DCB`), in its own map (`receiver_phase_bias_indices_`);
- satellite clock key (network): `"<sat>|SC"`, e.g. `"G05|SC"` — per satellite, *not* per signal; the trailing `|SC` discriminator gives the clock its own map (`sat_clock_indices_`);
- receiver code-bias key: `"<system>_IFB<band>"` or `"<system>_DCB"`; the per-slot GLONASS variant appends an FCN marker, `"GLONASS_IFB<band>_k<±slot>"` (e.g. `"GLONASS_IFB1_k-4"`);
- inter-system time-offset key: the bare system name (`"Galileo"`, …; GPS is the reference and never appears);
- double-difference ambiguity key (baseline): `"DD|<cons>|<sat>|<code>|<pivot>"`.

This single source of truth eliminates the off-by-one errors that plague hand-managed dynamic state vectors and makes the BSSD/DD S-transforms (Section 4) implementable as block operations on the relevant key sets.

Two further categories in the schema header are deliberately absent from the list above, because they are not rows of $\boldsymbol{x}$: the BSSD pivot absolute-seed lookup key (`"<cons>|<band>"`), and the shared spherical-harmonic VTEC coefficient keys (`"SH|<n>|<m>|<C|S>"`, with the second shell carried on the distinct `SH2` prefix), which name columns of the network estimator's *shared border* rather than `StateManager` states. Both nonetheless follow the same `make`/`parse` discipline, and both appear in the generated key-format table of the appendix — machine-derived from the schema header, and therefore the completeness reference against which the list above should be checked.

Ambiguity basis. By default the ambiguity block carries one per-band float ambiguity per tracked (satellite, signal) in metres. Setting `estimation.state_ambiguity_basis = "nl_wl"` switches the whole run to a narrow-lane / wide-lane parameterisation: per satellite a narrow-lane state (key `"<sat>_NL_<band>"`) and one wide-lane state per band pair (`"<sat>_WL_<a>_<b>"`, semantically $N_a - N_b$), both in *cycles* rather than metres so the wide-lane and LAMBDA arithmetic stays native to integers, with the wavelength scaling moved into the Jacobian. The two bases never coexist in a single state vector.

3 Observation equations

This section gives the full *undifferenced* observation model that the filter linearises around at each epoch. The PPP-RTK parameterisation is uncombined: per-signal pseudorange and carrier phase are consumed directly, without forming ionosphere-free or wide-lane combinations as a preprocessing step. Combinations (IF, MW, GF) are still computed for diagnostic and slip-detection purposes (the cycle-slip document; see `slip/MwSlipDetector`, `slip/GraphicSlipDetector`) but they do not enter the measurement update path. This holds whenever the uncombined parameterisation is active; the loader forces `uncombined_enabled=true` for the whole PPP-RTK family (user, network producer, network consumer) in `GnssConfig::loadProcessingSection_`; the same site forces it for the baseline double-difference modes, which process each signal independently (Section 4.3). Plain PPP mode leaves the flag at its default `false` and instead builds ionosphere-free combined observables (keys such as `C1W_C5X`) that *do* feed `MeasurementModel::buildJacobianRow` and the update; the sequel is written for the uncombined case.

3.1 Code (pseudorange) observation

For receiver r tracking satellite i on signal s, the observed pseudorange is

$$\boxed{P^i_{r,s} \;=\; \rho^i_r \;+\; c(\mathrm{d}t_r - \mathrm{d}t^i) \;+\; c\,\delta t_r^{(g(i))} \;+\; T^i_r \;+\; \mu_s\, I^i_{r,1} \;+\; B_{r,s} - B^i_s \;+\; \varepsilon^P_{r,i,s}.} \tag{4}$$

3.2 Carrier-phase observation

The corresponding carrier-phase observation, scaled to metres by λ_s, is

$$\boxed{\Phi^i_{r,s} \;=\; \rho^i_r \;+\; c(\mathrm{d}t_r - \mathrm{d}t^i) \;+\; c\,\delta t_r^{(g(i))} \;+\; T^i_r \;-\; \mu_s\, I^i_{r,1} \;+\; \lambda_s\, N^i_{r,s} \;+\; b_{r,s} - b^i_s \;+\; \varepsilon^\Phi_{r,i,s}.} \tag{5}$$

The defining structural difference between Equations (4) and (5) is the sign of the ionospheric term: $+\mu_s I$ on code, $-\mu_s I$ on phase. Three additional remarks:

1. The phase ambiguity $N^i_{r,s}$ is integer in principle but is contaminated by the receiver-plus-satellite phase biases $b_{r,s} - b^i_s$. In a *float* solution these biases are absorbed into $N^i_{r,s}$, leaving a non-integer state with no structural advantage over the code observation. Recovering integer-valued ambiguities therefore requires either external Fractional Cycle Bias (FCB) products or differencing (Section 4), which cancels common-mode bias terms.
2. The clock product $c\,\mathrm{d}t^i$ is read from the analysis-centre CLK file (post-processing) or refined from the broadcast clock by a streamed SSR correction (real-time, the SSR-corrections document) and treated as deterministic in the filter; any residual product error appears on the observation and is absorbed by the receiver clock state.
3. The system time offset $c\,\delta t_r^{(g(i))}$ is zero by construction when $g(i)$ is the reference constellation (default GPS). Together with the receiver clock $c\,\mathrm{d}t_r$ this means a multi-constellation run carries $|\mathcal{G}|$ clock-like states in total.

3.3 Geometric range and linearisation

The geometric range $\rho^i_r = \|\boldsymbol{r}^i - \boldsymbol{r}_r\|_2$ enters Equations (4) and (5) non-linearly through the receiver position $\boldsymbol{r}_r$. The filter linearises about the current state estimate $\hat{\boldsymbol{r}}_r$:

$$\rho^i_r \;=\; \hat{\rho}^i_r \;-\; (\boldsymbol{e}^i_{\hat{r}})^{\mathrm{T}}(\boldsymbol{r}_r - \hat{\boldsymbol{r}}_r) \;+\; \mathcal{O}\big(\|\boldsymbol{r}_r - \hat{\boldsymbol{r}}_r\|^2\big)\,, \tag{6}$$

with the unit line-of-sight

$$\boldsymbol{e}^i_{\hat{r}} \;:=\; \frac{\boldsymbol{r}^i - \hat{\boldsymbol{r}}_r}{\|\boldsymbol{r}^i - \hat{\boldsymbol{r}}_r\|_2}. \tag{7}$$

The Jacobian row $\boldsymbol{H}_{(i,s)}$ contributing to the design matrix at the position columns is therefore $-(\boldsymbol{e}^i_{\hat{r}})^{\mathrm{T}}$. For sub-metre residual targets this single linearisation per epoch is sufficient; the engine optionally relinearises within the epoch (`processing.relinearize_geometry`, which defaults to off in code and is enabled in the PPP, PPP-RTK, baseline, and real-time configs but left off in the six network-producer configs) to handle long initial position offsets.

The satellite position $\boldsymbol{r}^i$ is itself a function of the transmission time. Rather than iterate the emission time against the geometric range — which depends on $\boldsymbol{r}_r$ — `Geometry::computeEmissionState` solves it in a single step from the *pseudorange*: the time of flight is $\tau = P/c$, subtracted (with the satellite clock) from the receiver epoch. Because the receiver clock bias $\mathrm{d}t_r$ enters the receiver epoch as $+\mathrm{d}t_r$ and the pseudorange as $+c\,\mathrm{d}t_r$, it cancels exactly, leaving a residual $(I+T+\varepsilon)/c \approx 30$–$60$ ns of satellite displacement — sub-millimetre, below any PPP threshold. This deliberately avoids a range-based fixed point: iterating against ρ^i_r would re-introduce an uncancelled $\mathrm{d}t_r$ and drive a positive-feedback loop through the filter's receiver-clock state. The only second pass (the `model_corrections.refine_emission_time` block in `Geometry::computeSatelliteGeometry`) re-runs `computeEmissionState` on a cleaned pseudorange, having first subtracted the slant hydrostatic delay and the IF-combined ATX projection from the seed — a pseudorange-cleanliness refinement, not a range-versus-position iteration.

3.4 Stochastic model

Observation noise covariance is structured per epoch as a block diagonal matrix with entries

$$\sigma^2_{r,i,s} \;=\; \sigma^2_{0,\mathrm{type}(s)} \;\cdot\; w(\theta^i_r)^{-1} \;\cdot\; \xi(\mathrm{C/N_0 \text{ at } } (r,i,s)), \tag{8}$$

combining a per-type baseline ($\sigma_{0,\mathrm{code}}$, $\sigma_{0,\mathrm{phase}}$ from `filter.measurement_noise`), an elevation-dependent down-weighting factor $w(\theta^i_r)^{-1} = (\sin\theta^i_r)^{-2p_{\mathrm{type}}}$ whose exponent is *continuous* and *type-dependent* — code and Doppler rows take `filter.code_elevation_power`, carrier-phase rows `filter.phase_elevation_power` — with the weighting strategy itself selected by `filter.stochastic_model` (derived, together with the low-elevation guard that leaves σ^2_0 unamplified, in the stochastic-model section of the quality-control document), and an optional carrier-to-noise ratio scaling ξ parameterised by `snr_model_params`. Cross-correlations between observations are assumed zero in the undifferenced parameterisation; differencing introduces structured correlations handled in Section 4.

4 Differencing transformations

PPP-RTK and baseline modes do not consume the undifferenced observations of Equations (4) and (5) directly. Each forms a linear combination across satellites (BSSD), receivers (BRSD), or both (DD), absorbing common-mode terms into the cancellation and recovering integer-friendly residual structure. This section gives both the algebraic cancellations and the matrix-form S-transforms that map undifferenced parameters to differenced ones.

4.1 Between-satellite single difference (BSSD)

For each constellation g, choose a pivot satellite p_g (elected by `PivotManager::selectPivotCandidates` under the "sticky pivot" retention policy; the selection ordering is detailed in the *Pivot selection*

note below). For any non-pivot satellite $i \in \mathcal{S}_g \setminus \{p_g\}$ and signal s, define

$$\begin{aligned} \Delta P_{r,s}^{i,p_g} &:= P_{r,s}^{i} - P_{r,s}^{p_g}, \\ \Delta \Phi_{r,s}^{i,p_g} &:= \Phi_{r,s}^{i} - \Phi_{r,s}^{p_g}. \end{aligned} \tag{9}$$

Substituting Equations (4) and (5) and cancelling the receiver-side common terms ($c\,\mathrm{d}t_r$, $c\,\delta t_r^{(g)}$, $B_{r,s}$, $b_{r,s}$) yields

$$\Delta P_{r,s}^{i,p_g} = \Delta\rho_r^{i,p_g} - c\,\Delta \mathrm{d}t^{i,p_g} + \Delta T_r^{i,p_g} + \mu_s\,\Delta I_{r,1}^{i,p_g} - \Delta B_s^{i,p_g} + \Delta\varepsilon^{P}, \tag{10}$$

$$\Delta \Phi_{r,s}^{i,p_g} = \Delta\rho_r^{i,p_g} - c\,\Delta \mathrm{d}t^{i,p_g} + \Delta T_r^{i,p_g} - \mu_s\,\Delta I_{r,1}^{i,p_g} + \lambda_s\,\Delta N_{r,s}^{i,p_g} - \Delta b_s^{i,p_g} + \Delta\varepsilon^{\Phi}. \tag{11}$$

The cancellations are exact only when both observations enter the filter with the same modelling: same tropospheric mapping (which is why ΔT is small but not zero), same antenna PCV correction, and so on.

The receiver-bias cancellation of $B_{r,s}$ in Equation (10) holds for CDMA constellations, whose group delay is a single per-signal receiver term. GLONASS is the exception: its FDMA inter-frequency biases are per-FCN-slot states, so two satellites on different slots carry different receiver delays and the term does *not* cancel between them — a residual $B_{r,s}^{(k(i))} - B_{r,s}^{(k(p_g))}$ survives the single difference. Accordingly `BssdTransform` zeroes the receiver-DCB columns after differencing *except* the per-slot GLONASS IFB states (guarded by `StateKeys::isGlonassPerSlotIfbKey`), and `PivotManager::buildEpochMaps` drops GLONASS from the BSSD candidate maps altogether — not merely from pivot election — so with the flag off the constellation contributes no single-differenced row at all. GLONASS BSSD is default-OFF (`estimation.between_satellite_sd.enable_glonass`); when enabled, Equation (10) acquires the surviving inter-slot IFB difference as an additional estimated column.

Pivot selection. Election is not by elevation alone. The primary key is a *hard* lexicographic tier on phase-signal coverage: a full-coverage candidate outranks any partial-coverage candidate regardless of its score. Only within a coverage tier does the additive score $-\min\!\big(1, \sqrt{\sigma^2_{\hat{N},\max}}\big) + \mathrm{el}/90 - \frac{1}{2}\,\mathbb{1}[\mathrm{el} < 20^\circ]$ apply, in which float-ambiguity convergence $\sqrt{\sigma^2_{\hat{N},\max}}$ and elevation enter as co-equal terms (with a hard low-elevation guardrail below 20°). Layered on top is an ephemeris-freshness bar: a candidate whose broadcast $|\mathrm{obs} - \mathrm{toe}|$ exceeds its constellation's freshest qualified peer by more than `kPivotToeSpreadMaxS` (1200 s) is barred from *pivot* election (though still usable as a non-pivot), which prevents a stale-eph pivot from stepping every SD row it anchors. Candidates below `estimation.between_satellite_sd.pivot_min_elevation` are excluded outright. The baseline DD path scores differently: `DdPivotManager` ranks on $\mathrm{avg_snr} + 0.01 \cdot \mathrm{num_sigs}$ and does not use elevation.

Pivot transitions. A handover fires on two kinds of trigger. The visibility-driven one is simple — when the incumbent pivot leaves the visible set, p_g is re-elected from the remaining satellites — but the dominant one in practice is *event-driven*: the post-fit residual analyser calls `PivotManager::requestForcedHandover` when the pivot's *observation* slipped while its filter *state* is still intact (so the resting `isPivotDisturbed` check, which tests only that the pivot ambiguity sits at exactly $(0, 0)$, would not catch it). The forced request carries an exclusion set so the handover cannot bounce onto another offset satellite, a `retired_pivots_this_epoch_` backstop that terminates cascades, a same-epoch re-entry gate (`hasPendingForcedHandover`), and a re-inflation set for states that absorbed a

coherent pivot-side code bias (the information-conserving transform below cannot evict that offset on its own). The baseline DD path exposes the symmetric `DdPivotManager::requestForcedRerank`. This event response is load-bearing precisely because the pivot's states are frozen at zero variance (below): a stiff, time-linked zero-q quantity requires a wired jump response.

On election the float ambiguities $\{\Delta N_{r,s}^{i,p_g^{\mathrm{old}}}\}_{i,s}$ must be re-expressed relative to p_g^{new}. Where the new pivot already carries an established ambiguity in the group, the engine applies the linear S-transformation $\boldsymbol{x}^{\mathrm{new}} = \boldsymbol{S}\,\boldsymbol{x}^{\mathrm{old}}$ with $\boldsymbol{S}$ chosen so that $\Delta N_{r,s}^{p_g^{\mathrm{new}},p_g^{\mathrm{old}}}$ becomes the new reference (zero), preserving the integer information accumulated since the last fix; the covariance transforms as $\boldsymbol{P}^{\mathrm{new}} = \boldsymbol{S}\,\boldsymbol{P}^{\mathrm{old}}\,\boldsymbol{S}^{\mathrm{T}}$. When it cannot — the new pivot has no existing ambiguity in the (cons, code) group (`DdPivotManager`, logged as "reset-and-reseed"), or a disturbed pivot has no healthy replacement in a single-satellite constellation (`PivotManager::handleDisturbedPivot` re-zeroes $\boldsymbol{x}$ and the corresponding rows/columns of $\boldsymbol{P}$ in place) — the transform is skipped and the stale rows are discarded and re-seeded. In the uncombined PPP-RTK parameterisation the same $\boldsymbol{S}\,\boldsymbol{P}\,\boldsymbol{S}^{\mathrm{T}}$ re-reference is applied not only to the ambiguity block but to *three* state groups — `reReferenceAmbiguities`, `reReferenceIonosphere`, and `reReferenceSatBiases` — which is why `BssdTransform` zeroes the pivot's ionosphere and sat-bias $\boldsymbol{H}$-columns after differencing, and why $\Delta I_{r,1}^{i,p_g}$ in Equations (10) and (11) is itself a pivot-referenced SD quantity, not an absolute slant delay. The S-basis constraint is realised operationally by freezing the pivot's ambiguity, ionosphere, and sat-bias covariances to exactly zero (and suppressing their process noise) — the very condition `isPivotDisturbed` tests against.

`PivotManager::getCleanHandovers` classifies each transition as lossless-invertible (*clean*) or discontinuous (disturbed/forced/re-seeded), so the RTS smoother can inverse-reference across a clean handover rather than force a segment boundary; the losslessness claim above holds only for the clean subset. These are two distinct code paths sharing the algebra above: in PPP-RTK the pivot re-election and the "sticky-pivot" retention policy live in `filter/PivotManager`, whereas the baseline DD path performs the analogous re-reference through the ambiguity-rename S-transform in `filter/DdPivotManager`, governed by the `baseline.enable_pivot_handover` flag.

4.2 Between-receiver single difference (BRSD)

In baseline modes there is at least one rover r_{rov} and exactly one fixed-coordinate reference r_{ref}. The BRSD operator forms

$$\nabla P^{i}_{r_{\mathrm{rov}}r_{\mathrm{ref}},s} \;:=\; P^{i}_{r_{\mathrm{rov}},s} - P^{i}_{r_{\mathrm{ref}},s}, \tag{12}$$

and analogously for $\nabla\Phi$. The cancellations produced by $\nabla\cdot$ are dual to those produced by $\Delta\cdot$: BRSD cancels *satellite*-side common terms ($c\,\mathrm{d}t^i$, B_s^i, b_s^i) but preserves receiver clock and biases of *both* stations. No state-space estimation of satellite biases is required when ∇ is applied — they disappear before reaching the filter.

4.3 Double difference (DD)

The composition $\nabla\Delta\cdot = \Delta\big(\nabla\,\cdot\big)$ is order-independent up to noise correlation: $\Delta\big(\nabla\,\cdot\big) = \nabla\big(\Delta\,\cdot\big)$. Applying it to the carrier-phase model and collecting common-mode cancellations:

$$\nabla\Delta\Phi^{i,p_g}_{r_{\mathrm{rov}}r_{\mathrm{ref}},s} \;=\; \nabla\Delta\rho^{i,p_g}_{r_{\mathrm{rov}}r_{\mathrm{ref}}} + \nabla\Delta T^{i,p_g}_{r_{\mathrm{rov}}r_{\mathrm{ref}}} - \mu_s\,\nabla\Delta I^{i,p_g}_{r_{\mathrm{rov}}r_{\mathrm{ref}},1} + \lambda_s\,\nabla\Delta N^{i,p_g}_{r_{\mathrm{rov}}r_{\mathrm{ref}},s} + \nabla\Delta\varepsilon^{\Phi}. \tag{13}$$

What survives in Equation (13) is precisely the geometry term (linear in the rover position perturbation), the differential atmospheric delays, and the integer DD ambiguity — the structural form that makes carrier-phase RTK possible.

Short vs. long baselines. On short baselines ($\lesssim 10\,\mathrm{km}$) the differential atmospheric terms in Equation (13) are negligible at the millimetre level and need not be modelled. As baseline length grows, the differential ionosphere $\nabla\Delta I$ scales linearly with separation and saturates at typically 1 m for separations of a few hundred kilometres, swamping the carrier wavelength. Two mitigations are available in `GnssProcessor`:

- `baseline.use_iono_free` forms the dual-frequency ionosphere-free combination at the DD level, eliminating the first-order ionospheric term at the cost of a non-integer ambiguity (AR is automatically excluded for IF rows) and $\sim 3\times$ amplified code noise;
- `baseline.estimate_differential_tropo` adds a single $\Delta\mathrm{ZWD}_{r_{\mathrm{rov}}r_{\mathrm{ref}}}$ state with slow random-walk dynamics, capturing the residual differential wet delay that survives the BRSD on baselines with significant altitude difference.

4.4 S-transform structure

Both BSSD and DD admit a matrix-form representation as $\boldsymbol{x}_{\mathrm{diff}} = \boldsymbol{S}\,\boldsymbol{x}_{\mathrm{undiff}}$ with $\boldsymbol{S}$ acting as a block design transform. For BSSD restricted to the ambiguity block of one constellation g on one signal s, with $|\mathcal{S}_g|$ tracked sats, $\boldsymbol{S}$ is a $(|\mathcal{S}_g|-1)\times|\mathcal{S}_g|$ matrix of the form

$$\boldsymbol{S}_{\mathrm{BSSD}} = \begin{bmatrix} 1 & 0 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots & & & \\ 0 & 0 & \cdots & 0 & -1 & 0 & \cdots & 1 \end{bmatrix}, \tag{14}$$

where the column of -1's sits at the index of the pivot p_g. The DD operator is the *composition* of the two single-difference operators applied in sequence, $\boldsymbol{S}_{\mathrm{DD}} = \boldsymbol{S}_{\mathrm{BSSD}}\,\boldsymbol{S}_{\mathrm{BRSD}}$, realised in code as `BrsdTransform::apply` followed by `BssdTransform::differenceBrsd` (`include/measurement/BssdTransform.hpp`). Only on a fully populated satellite $\times$ receiver observation grid does this composition coincide with the Kronecker product $\mathbf{D}_{\mathrm{sat}} \otimes \mathbf{D}_{\mathrm{rx}}$ of the between-satellite and between-receiver differencing matrices acting on the vectorised observations — a useful mental model, but not what the non-rectangular real-data case computes. The transformed covariance is $\boldsymbol{P}_{\mathrm{diff}} = \boldsymbol{S}\,\boldsymbol{P}_{\mathrm{undiff}}\,\boldsymbol{S}^{\mathrm{T}}$, a generally dense matrix even when $\boldsymbol{P}_{\mathrm{undiff}}$ is diagonal (because each differenced row borrows from the same pivot column). This dense correlation structure is the algebraic reason DD-based filters cannot be naively implemented as independent per-pair scalar updates — the correctly correlated joint update is what unlocks integer ambiguity recovery via the LAMBDA decorrelation. The general theory and a thorough catalogue of admissible S-transforms is given in Teunissen [4] and in the rank-defect treatment of Jonge [1].

Implementation notes. Two facts about how `GnssProcessor` realises the transforms above are worth stating explicitly. First, the chosen S-system is never materialised as an explicit minimum-constraint $\boldsymbol{S}$-basis matrix; it is realised *implicitly* through the pivot-differencing operator (`measurement/BssdTransform`), a faithful, S-basis-invariant realisation of the uncombined parameterisation of Odijk [2]. The three regimes — undifferenced, BSSD, and DD — are dispatched polymorphically by a `DifferencingMode` strategy in `measurement/MeasurementBuilder`, not by inline config branches. Second, integer recovery on a DD row requires that all four contributing observations share the same RINEX signal code. The row is marked `integer_capable` only when all three conjuncts hold: `!brsd_aliased` on *both* the i_i and pivot BRSD entries (rover and reference agree on each, i.e. no cross-attribute alias) *and* the rover uses the same code on i_i as on the pivot

($\texttt{rov_code}_i = \texttt{rov_code}_{\text{piv}}$). The third conjunct is independent of the `brsd_aliased` flags: a row can be un-aliased on both entries yet still carry a non-integer DCB/IFB residual if the rover mixes codes across i_i and the pivot. Rows failing any conjunct are carried as float and excluded from ambiguity resolution.

References

[1] P. J. de Jonge. "A processing strategy for the application of the GPS in networks". PhD thesis. Delft, The Netherlands: Delft University of Technology, 1998.

[2] D. Odijk. "Fast precise GPS positioning in the presence of ionospheric delays". In: *Publications on Geodesy* 52 (2003).

[3] J. Saastamoinen. "Atmospheric correction for the troposphere and stratosphere in radio ranging of satellites". In: *The Use of Artificial Satellites for Geodesy*. Ed. by S. W. Henriksen, A. Mancini, and B. H. Chovitz. Vol. 15. Geophysical Monograph Series. Washington, D.C.: American Geophysical Union, 1972, pp. 247–251. DOI: 10.1029/GM015p0247.

[4] P. J. G. Teunissen. "Generalized inverses, adjustment, the datum problem and S-transformations". In: *Optimization and Design of Geodetic Networks* (1985), pp. 11–55.

[5] P. J. G. Teunissen. "The least-squares ambiguity decorrelation adjustment: a method for fast GPS integer ambiguity estimation". In: *Journal of Geodesy* 70.1–2 (1995), pp. 65–82. DOI: 10.1007/BF00863419.